

606024



Test Booklet No.

ENGLISH

Test Booklet Code

08

# NEET RANKERS

## TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

### Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
- The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. 50 questions in each subject are divided into **two Sections (A and B)** as per details given below:
  - Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
  - Section B** shall consist of **15 (Fifteen)** questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each subject.

**Candidates are advised to read all 15 questions in each subject of Section B** before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated.**
- Each question carries 4 marks. For each correct response, the candidate will get **4 marks**. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
- Use **Blue/Black Ball Point Pen** only for writing particulars on this page/markings responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The **CODE** for this Booklet is **08**. **Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice**. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): \_\_\_\_\_

Roll Number : In figures \_\_\_\_\_

: In words \_\_\_\_\_

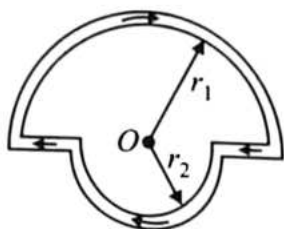
Centre of Examination (in Capitals): \_\_\_\_\_

Candidate's Signature: \_\_\_\_\_ Invigilator's Signature: \_\_\_\_\_

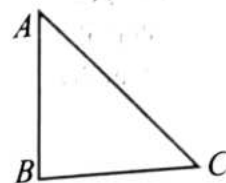
Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

1. In the figure shown, there are two semicircles of radii  $r_1$  and  $r_2$  in which a current  $i$  is flowing as shown in figure. The magnetic induction at the centre  $O$  will be:



- (1)  $\frac{\mu_0 i}{r}(r_1 + r_2)$  (2)  $\frac{\mu_0 i}{4}(r_1 - r_2)$   
 (3)  $\frac{\mu_0 i}{4}\left(\frac{r_1 + r_2}{r_1 r_2}\right)$  (4)  $\frac{\mu_0 i}{4}\left(\frac{r_2 - r_1}{r_1 r_2}\right)$
2. A monoatomic gas at a pressure  $P$ , having a volume  $V$  expands isothermally to a volume  $2V$  and then adiabatically to a volume  $16V$ . The final pressure of the gas is : (take  $\gamma = \frac{5}{3}$ )
- (1)  $64P$  (2)  $32P$   
 (3)  $\frac{P}{64}$  (4)  $16P$
3. In Young's double slit experiment, when two light waves form third order minima, they have:
- (1) phase difference of  $3\pi$ .  
 (2) phase difference of  $\frac{5\pi}{2}$ .  
 (3) path difference of  $3\lambda$ .  
 (4) path difference of  $\frac{5\lambda}{2}$ .
4. A current carrying closed loop in the form of a right angle isosceles triangle  $ABC$  is placed in a uniform magnetic field acting along  $AB$ . If the magnetic force on the arm  $BC$  is  $\vec{F}$ , the force on the arm  $AC$  is:



- (1)  $-\sqrt{2}\vec{F}$  (2)  $-\vec{F}$   
 (3)  $\vec{F}$  (4)  $\sqrt{2}\vec{F}$
5. Two charges of equal magnitudes and at a distance  $r$  exert a force  $F$  on each other. If the charges are halved and distance between them is doubled, then the new force acting on each charge is:
- (1)  $F/8$  (2)  $F/4$   
 (3)  $4F$  (4)  $F/16$
6. During the first 18 min of a 60 min trip, a car has an average speed of  $11 \text{ ms}^{-1}$ . What should be the average speed for remaining 42 min so that car is having an average speed of  $21 \text{ ms}^{-1}$  for the entire trip?
- (1)  $25.3 \text{ ms}^{-1}$  (2)  $29.2 \text{ ms}^{-1}$   
 (3)  $31 \text{ ms}^{-1}$  (4)  $35.6 \text{ ms}^{-1}$
7. By what factor the rms velocity will change, if the temperature is raised from  $27^\circ\text{C}$  to  $327^\circ\text{C}$ ?
- (1)  $\sqrt{2}$  (2) 2  
 (3)  $2\sqrt{2}$  (4) 1
8. Dimensions of  $\epsilon_0 \frac{d\phi_E}{dt}$  are same as that of:
- (1) charge. (2) potential.  
 (3) capacitance. (4) current.
9. A particle undergoing SHM has the equation  $x = A\sin(2\omega t + \phi)$ , where  $x$  represents the displacement of the particle. The kinetic energy oscillates with time period:
- (1)  $\frac{2\pi}{\omega}$  (2)  $\frac{\pi}{\omega}$   
 (3)  $\frac{4\pi}{\omega}$  (4)  $\frac{\pi}{2\omega}$

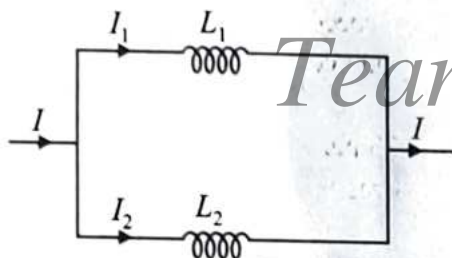
10. A particle with the position vector  $r$  has linear momentum  $p$ . Which of the following statements is **true** in respect of its angular momentum  $L$  about the origin?

(1)  $L$  acts along  $p$ .  
 (2)  $L$  acts along  $r$ .  
 (3)  $L$  is maximum when  $p$  and  $r$  are parallel.  
 (4)  $L$  is maximum when  $p$  is perpendicular to  $r$ .

11. The S.I. unit of magnetic permeability is:

(1)  $\text{Am}^{-1}$ .  
 (2)  $\text{A-m}$ .  
 (3)  $\text{Henry m}^{-1}$ .  
 (4) no unit, it is a dimensionless number.

12. Two inductors  $L_1$  and  $L_2$  are connected in parallel and a time varying current flows as shown in figure. Then the ratio of currents  $I_1/I_2$  at any time  $t$  is:



(1)  $L_1/L_2$   
 (2)  $L_2/L_1$   
 (3)  $\frac{L_1}{(I_1 + I_2)^2}$   
 (4)  $\frac{L_2}{(I_1 + I_2)^2}$

13. A gate has the following truth table

| $P$ | $Q$ | $R$ |
|-----|-----|-----|
| 1   | 1   | 1   |
| 1   | 0   | 0   |
| 0   | 1   | 0   |
| 0   | 0   | 0   |

The gate is:

(1) NOR (2) OR  
 (3) NAND (4) AND

14. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

**Assertion A:** Pressure has the dimensions of energy density.

**Reason R:** Energy density

$$= \frac{\text{energy}}{\text{volume}} = \frac{[ML^2T^{-2}]}{[L^3]} = [ML^{-1}T^{-2}] = \text{pressure.}$$

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.  
 (2) A is false but R is true.  
 (3) Both A and R are true and R is the correct explanation of A.  
 (4) Both A and R are true but R is NOT the correct explanation of A.

15. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

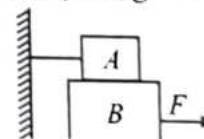
**Assertion A:** Adiabatic bulk modulus of an ideal gas is more than its isothermal bulk modulus.

**Reason R:** Both the moduli are proportional to the pressure of gas for same conditions.

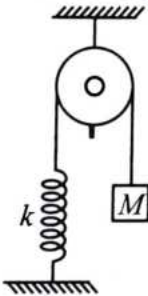
In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.  
 (2) A is false but R is true.  
 (3) Both A and R are true and R is the correct explanation of A.  
 (4) Both A and R are true but R is NOT the correct explanation of A.

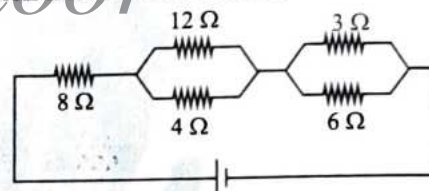
16. A block A of mass 100 kg rests on another block B of mass 200 kg and is tied to a wall as shown in the figure. The coefficient of friction between A and B is 0.2 and that between B and the ground is 0.3. The minimum force  $F$  required to move the block B is: (take,  $g = 10 \text{ ms}^{-2}$ )



(1) 900 N (2) 200 N  
 (3) 1100 N (4) 700 N

17. If the work function of potassium is 2 eV, then its photoelectric threshold wavelength is:  
 (1) 310 nm (2) 620 nm  
 (3) 6200 nm (4) 3100 nm
18. Eight oil drops of same size are charged to a potential of 50 V each. These oil drops are merged into one single large drop. What will be the potential of the large drop?  
 (1) 50 V (2) 100 V  
 (3) 200 V (4) 400 V
19. A particle of mass  $m$  and charge  $Q$  moving with a velocity  $\vec{v}$  enters a region of uniform magnetic field of induction  $\vec{B}$ . Then, its path in the region is:  
 (1) always circular.  
 (2) circular, if  $\vec{v} \times \vec{B} = 0$ .  
 (3) circular, if  $\vec{v} \cdot \vec{B} = 0$ .  
 (4) always straight line.
20. A mass  $M = 5$  kg is attached to a spring as shown in the figure and held in position so that the spring remains unstretched. The spring constant is  $200 \text{ Nm}^{-1}$ . The mass  $M$  is then released and begins to undergo small oscillations. The amplitude of oscillation is:  
  
 (1) 0.5 m (2) 0.25 m  
 (3) 0.2 m (4) 0.1 m
21. The electromagnetic wave with shortest wavelength among the following is:  
 (1) UV-rays. (2) X-rays.  
 (3)  $\gamma$ -rays. (4) microwaves.

22. The orbital velocity of an artificial satellite in a circular orbit just above the earth's surface is  $v$ . For a satellite orbiting at an altitude of half the earth's radius the orbital velocity is:  
 (1)  $\frac{3}{2}v$  (2)  $\sqrt{\frac{3}{2}}v$   
 (3)  $\sqrt{\frac{2}{3}}v$  (4)  $\frac{2}{3}v$
23. In position A kinetic energy of a particle is 60 J and potential energy is -20 J. In position B, kinetic energy is 100 J and potential energy is 40 J. Then, in moving the particle from A to B:  
 (1) work done by conservative forces is 50 J.  
 (2) work done by external forces is 40 J.  
 (3) net work done by all the forces is 40 J.  
 (4) net work done by all the forces is 100 J.
24. In the circuit diagram shown in figure, potential difference across  $3 \Omega$  resistance is 20 V. Then, match List-I with List-II:



| List-I                                                | List-II   |
|-------------------------------------------------------|-----------|
| A. Potential difference across $6 \Omega$ resistance  | I. 30 V   |
| B. Potential difference across $4 \Omega$ resistance  | II. 40 V  |
| C. Potential difference across $12 \Omega$ resistance | III. 20 V |
| D. Potential difference across $8 \Omega$ resistance  | IV. 80 V  |

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II  
 (2) A-I, B-III, C-III, D-IV  
 (3) A-III, B-I, C-I, D-IV  
 (4) A-III, B-II, C-IV, D-I

25. A particle of mass  $m$  is projected with a velocity  $v$  making an angle of  $45^\circ$  with the horizontal. The magnitude of angular momentum of projectile about the point of projection when the particle is at its maximum height  $h$  is:
- (1) zero (2)  $\frac{mvh}{\sqrt{2}}$   
 (3)  $mvh$  (4)  $\sqrt{2}mvh$
26. The orbital frequency of an electron in hydrogen like atom is proportional to:
- (1)  $n^3$  (2)  $n^{-3}$   
 (3)  $n$  (4)  $n^0$
27. In order to obtain a real image of magnification 2 using converging lens of focal length 20 cm, where should a real object be placed?
- (1) 50 cm (2) 30 cm  
 (3) -50 cm (4) -30 cm
28. The transformation ratio in the step-up transformer is:
- (1) 1.  
 (2) greater than one.  
 (3) less than one.  
 (4) the ratio greater or less than one depends on the other factors.
29. A point object of mass  $m$  is kept at  $(a, 0)$  on the  $x$ -axis. What mass should be kept at  $(-3a, 0)$ , so that centre of mass lie at the origin?
- (1)  $m$  (2)  $2m$   
 (3)  $\frac{m}{3}$  (4)  $3m$
30. A string of length 0.1 m cannot bear a tension more than 100 N. It is tied to a body of mass 100 g and rotated in a horizontal circle. The maximum angular velocity can be:
- (1) 100 rad  $s^{-1}$   
 (2) 1000 rad  $s^{-1}$   
 (3) 10000  $s^{-1}$   
 (4) 0.1 rad  $s^{-1}$
31. Pick out the correct statement from the following.
- (1)  $\alpha$ -particles are not deflected by electric and magnetic fields.  
 (2)  $\beta$ -particle has less penetrating power than  $\alpha$ -particle.  
 (3)  $\gamma$ -rays has the least ionising power than  $\alpha$  and  $\beta$ -particles.  
 (4)  $\gamma$ -rays has the least penetrating power than  $\alpha$  and  $\beta$ -rays.
32. The escape velocity of a particle of mass  $m$  varies as:
- (1)  $m^2$  (2)  $m$   
 (3)  $m^0$  (4)  $m^{-1}$
33. The electric field at a point on the equatorial plane at a distance  $r$  from the centre of a dipole having dipole moment  $p$  is given by: ( $r \gg$  separation of two charges forming the dipole,  $\epsilon_0 =$  permittivity of free space)
- (1)  $E = \frac{p}{4\pi\epsilon_0 r^3}$  (2)  $E = \frac{2p}{4\pi\epsilon_0 r^3}$   
 (3)  $E = -\frac{p}{4\pi\epsilon_0 r^2}$  (4)  $E = -\frac{p}{4\pi\epsilon_0 r^3}$
34. The self-inductance of a coil having 500 turns is 50 mH. The magnetic flux through the cross-sectional area of the coil while current through it is 8 mA is found to be:
- (1)  $4 \times 10^{-4}$  Wb (2) 0.04 Wb  
 (3) 4  $\mu$ Wb (4) 40 mWb
35. The upper half of an inclined plane of inclination  $\theta$  is perfectly smooth while lower half is rough. A block starting from rest at the top of the plane will again come to rest at the bottom, if the coefficient of friction between the block and lower half of the plane is given by:
- (1)  $\mu = \frac{1}{\tan \theta}$  (2)  $\mu = \frac{2}{\tan \theta}$   
 (3)  $\mu = 2 \tan \theta$  (4)  $\mu = \tan \theta$

Physics : Section-B (Q. No. 36 to 50)

36. Light waves can be polarised as they are:  
 (1) transverse. (2) of high frequency.  
 (3) longitudinal. (4) reflected.
37. A block of mass  $m$  moving at a velocity  $v$  collides with another block of mass  $2m$  at rest. The lighter block comes to rest after collision. Find the coefficient of restitution.  
 (1)  $\frac{1}{2}$  (2) 1  
 (3)  $\frac{1}{3}$  (4)  $\frac{1}{4}$
38. Two resistors  $R$  and  $2R$  are connected in series in an electric circuit. The thermal energy developed in  $R$  and  $2R$  are in the ratio:  
 (1) 1 : 2 (2) 2 : 1  
 (3) 1 : 4 (4) 4 : 1
39. An air column, closed at one end and open at the other, resonates with a tuning fork when the smallest length of the column is 50 cm. The next larger length of the column resonating with the same tuning fork is:  
 (1) 100 cm. (2) 150 cm.  
 (3) 200 cm. (4) 66.7 cm.
40. If the velocity of an electron is doubled, its de-Broglie frequency will:  
 (1) be halved. (2) remain same.  
 (3) be doubled. (4) become four times.
41. A cuboid has volume  $V = l \times 2l \times 3l$ , where  $l$  is the length of one side. If the relative percentage error in the measurement of  $l$  is 1%, then the relative percentage error in measurement of  $V$  is:  
 (1) 18% (2) 6%  
 (3) 3% (4) 1%
42. Three capacitors each of capacity  $4 \mu\text{F}$  are to be connected in such a way that the effective capacitance is  $6 \mu\text{F}$ . This can be done by:

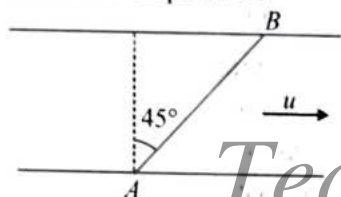
- (1) connecting them in parallel.  
 (2) connecting two in series and one in parallel.  
 (3) connecting two in parallel and one in series.  
 (4) connecting all of them in series.
43. The position of a particle moving in the  $xy$ -plane at any time  $t$  is defined as  $x = (3t^2 - 6t)$ ,  $y = (t^2 - 2t)$  m. Choose the **correct** statement(s) for the moving particle from the given below:  
 A. At  $t = 0$  second the acceleration of the particle is zero.  
 B. At  $t = 0$  second the velocity of the particle is zero.  
 C. At  $t = 1$  second the velocity of the particle is zero.  
 (1) A only (2) A, C only  
 (3) C only (4) A, B only
44. Kinetic energy of a particle moving in a straight line varies with time  $t$  as  $K = 4t^2$ . The force acting on the particle:  
 (1) is constant.  
 (2) is increasing.  
 (3) is decreasing.  
 (4) first increases and then decreases.
45. In an AC circuit, an alternating voltage  $e = 200\sqrt{2} \sin 100t$  V is connected to a capacitor of capacity  $1 \mu\text{F}$ . The rms value of the current in the circuit is:  
 (1) 100 mA. (2) 200 mA.  
 (3) 20 mA. (4) 10 mA.
46. When light rays are incident on a prism at an angle of  $45^\circ$ , the minimum deviation is obtained. If refractive index of the material of prism is  $\sqrt{2}$ , then the angle of prism will be:  
 (1)  $30^\circ$  (2)  $75^\circ$   
 (3)  $90^\circ$  (4)  $60^\circ$
47. Charge of  $2C$  is placed at the centre of a cube. What is the electric flux passing through one face?

- (1)  $\frac{1}{3\epsilon_0}$  (2)  $\left(\frac{1}{4}\right)\epsilon_0$   
 (3)  $\frac{2}{\epsilon_0}$  (4)  $\frac{3}{\epsilon_0}$

48. On increasing the reverse bias to a large value in a p-n junction diode, current:

- (1) increases slowly.  
 (2) remains fixed.  
 (3) suddenly increases.  
 (4) decreases slowly

49. A man wants to reach point B on the opposite bank of a river flowing at a speed as shown in figure. With what minimum speed and in which direction should the man swim relative to water so that he can reach point B?



- (1)  $u$ ,  $45^\circ$  North-West  
 (2)  $u$ ,  $45^\circ$  North-East  
 (3)  $\frac{u}{\sqrt{2}}$ ,  $45^\circ$  North-West  
 (4)  $\frac{u}{\sqrt{2}}$ ,  $45^\circ$  North-East

50. A rod AB is 1 m long. The temperature of its one end A is maintained at  $100^\circ\text{C}$  and other end B at  $10^\circ\text{C}$ , the temperature at a distance of 60 cm from point B is:

- (1)  $64^\circ\text{C}$  (2)  $36^\circ\text{C}$   
 (3)  $46^\circ\text{C}$  (4)  $72^\circ\text{C}$

**Chemistry : Section-A (Q. No. 51 to 85)**

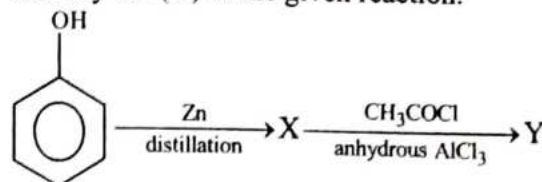
51. IUPAC name for given structure is

- (1) 2-Ethylbutan-2-ol  
 (2) Diethylmethylmethanol  
 (3) 1-Methyl-1-ethylpropanol  
 (4) 3-Methylpentan-3-ol

52. The oxidation number of 'S' in  $\text{H}_2\text{SO}_5$  is:

- (1) 8 (2) 6  
 (3) -2 (4) 4

53. Identify the (Y) in the given reaction:



- (1) Benzophenone  
 (2) Acetophenone  
 (3) Diphenyl  
 (4) Methyl salicylate

54. The product formed in the reaction when primary amine and aldehyde, reacts in the mild acidic medium is;

- (1)  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OH}$   
 (2)  $\text{R}-\text{ONO}$   
 (3)  $\text{R}'-\text{CH}=\text{N}-\text{R}$   
 (4)  $\text{R}-\text{NO}_2$

55. A gas is found to have the formula  $(\text{CO})_x$ . its vapour density is 70 the value of x must be;

- (1) 7 (2) 4  
 (3) 5 (4) 6

56. Match List-I with List-II.

| List-I<br>(Compound) | List-II<br>(Shape/Geometry) |
|----------------------|-----------------------------|
| (A) $\text{XeF}_4$   | (I) Trigonal bipyramidal    |
| (B) $\text{ClF}_3$   | (II) Square planar          |
| (C) $\text{PCl}_5$   | (III) Trigonal pyramidal    |
| (D) $\text{NH}_3$    | (IV) T-shape                |

Choose the correct answer from the options given below:

- (1) A-II, B-I, C-III, D-IV  
 (2) A-II, B-IV, C-I, D-III  
 (3) A-III, B-IV, C-II, D-I  
 (4) A-IV, B-III, C-II, D-I

57. Which of the following transition metals does not show variable valency in d-block?

- (1) Mn (2) Sc  
(3) Fe (4) V

58. Match List-I with List-II.

| List-I<br>(Value of $\Delta H$ & $\Delta S$ )  | List-II<br>(Feasibility of process)           |
|------------------------------------------------|-----------------------------------------------|
| (A) $\Delta H = -ve$ (I)<br>$\Delta S = -ve$   | Process is at spontaneous at all temperature  |
| (B) $\Delta H = +ve$ (II)<br>$\Delta S = +ve$  | Process is non spontaneous at all temperature |
| (C) $\Delta H = -ve$ (III)<br>$\Delta S = +ve$ | Process spontaneous at high temperature       |
| (D) $\Delta H = +ve$ (IV)<br>$\Delta S = -ve$  | Process spontaneous at low temperature        |

Choose the correct answer from the options given below:

- (1) A-I, B-II, C-III, D-IV  
(2) A-IV, B-III, C-I, D-II  
(3) A-II, B-IV, C-I, D-III  
(4) A-III, B-IV, C-I, D-II

59. The  $E_a$  for an exothermic reaction  $A \rightarrow B$  is 80 kJ mol<sup>-1</sup>. Heat of reaction is 20 kJ mol<sup>-1</sup>.  $E_a$  for the reaction  $B \rightarrow A$  will be;

- (1) 100 kJ/mol  
(2) 60 kJ/mol  
(3) 50 kJ/mol  
(4) 70 kJ/mol

60. Match List-I with List-II.

| List-I<br>(Vitamin)        | List-II<br>(Deficiency disease) |
|----------------------------|---------------------------------|
| (A) Vitamin A              | (I) Beri beri                   |
| (B) Vitamin B <sub>1</sub> | (II) Bleeding gums              |
| (C) Vitamin B <sub>6</sub> | (III) Xerophthalmia             |
| (D) Vitamin C              | (IV) Convulsions                |

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV  
(2) A-IV, B-III, C-II, D-I  
(3) A-II, B-III, C-IV, D-I  
(4) A-III, B-I, C-IV, D-II

61. Spin only magnetic moment is same for which of the following ions.

- A. Co<sup>2+</sup> B. Cu<sup>2+</sup>  
C. V<sup>2+</sup> D. Zn<sup>2+</sup>

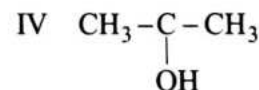
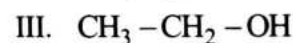
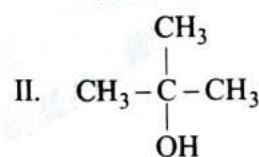
Choose the most appropriate answer from the given options below:

- (1) A and B only  
(2) A and C only  
(3) A, B and C only  
(4) A, B, C and D

62. The correct increasing order of acidic strength is;

- (1) CH<sub>3</sub>COOH < HCOOH < C<sub>6</sub>H<sub>5</sub>COOH  
(2) HCOOH < C<sub>6</sub>H<sub>5</sub>COOH < CH<sub>3</sub>OOH  
(3) C<sub>6</sub>H<sub>5</sub>COOH < CH<sub>3</sub>COOH < HCOOH  
(4) CH<sub>3</sub>COOH < C<sub>6</sub>H<sub>5</sub>COOH < HCOOH

63. The correct reactivity order of alcohols towards HBr will be:



- (1) II > I > III > IV (2) IV > III > II > I  
(3) III > IV > I > II (4) II > IV > III > I

64. Given below are two statements:

**Statement I:** The primary valences are ionisable in coordination compounds.



**Statement II:** The secondary valences are non-ionisable in coordination compounds.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

65. Given below are two statements:

**Statement I:** Axial P-Cl bonds are longer than equatorial P-Cl bonds in  $\text{PCl}_5$ .

**Statement II:** In solid state,  $\text{PCl}_5$  exists as  $[\text{PCl}_4]^-$  and  $[\text{PCl}_6]^+$  ions.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

66. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R: **Assertion A:** In Rutherford experiment, most of  $\alpha$ -particles passed through the gold foil undeflected.

**Reason R:** Maximum part of atom is empty.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

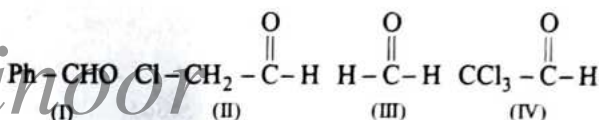
67. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R: **Assertion A:**  $\text{S}_{\text{N}}1$  reactions are carried out in polar protic solvents.

**Reason R:** Rate of reaction in  $\text{S}_{\text{N}}1$  depends upon the concentration of substrate.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is not the correct explanation of A.

68. Arrange the compounds in order of decreasing reactivity nucleophilic addition reaction:



- (1)  $\text{IV} > \text{II} > \text{III} > \text{I}$
- (2)  $\text{I} > \text{II} > \text{III} > \text{IV}$
- (3)  $\text{II} > \text{III} > \text{I} > \text{IV}$
- (4)  $\text{II} > \text{I} > \text{III} > \text{IV}$

69. Conjugate base of hydrazoic acid ( $\text{HN}_3$ ) is:

- (1)  $\text{H}_3^-$
- (2)  $\text{N}_3^-$
- (3)  $\text{N}^{3-}$
- (4)  $\text{N}_2^-$

70. Match List-I with List-II.

| List-I<br>(Molecule) | List-II<br>(Number and types of bond between two carbon atoms) |
|----------------------|----------------------------------------------------------------|
| (A) Propene          | (I) Six $\sigma$ -bonds, three $\pi$ -bonds                    |
| (B) Propyne          | (II) Only one $\sigma$ -bond                                   |
| (C) Benzene          | (III) Two $\sigma$ -bond and one $\pi$ -bond                   |
| (D) Ethane           | (IV) Two $\sigma$ -bond and two $\pi$ -bond                    |

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-I, D-IV  
 (2) A-IV, B-I, C-II, D-III  
 (3) A-III, B-IV, C-I, D-II  
 (4) A-I, B-II, C-III, D-IV

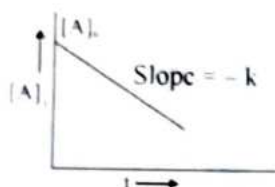
71. Which one of the following is monobasic acid?  
 (1)  $\text{H}_3\text{PO}_2$  (2)  $\text{H}_3\text{PO}_3$   
 (3)  $\text{H}_3\text{PO}_4$  (4)  $\text{H}_4\text{P}_2\text{O}_7$

72. Does Le chatelier's principle predict a change of equilibrium concentration for the following reaction if the gas mixture is compressed:



- (1) Yes, backward reaction is favoured  
 (2) Yes, forward reaction is favoured  
 (3) No change in equilibrium.  
 (4) Initially forward after some time backward.

73. For a chemical reaction in concentration  $[A]$  vs time  $(t)$  plot is given below, the order of reaction is:



- (1) Zero order (2) First order  
 (3) Second order (4) Third order

74. In the reaction  $\text{A} + 2\text{B} \rightleftharpoons 2\text{C}$ , if 2 moles of A, 3 moles of B and 2 moles of C are placed in a 2 L flask. The equilibrium concentration of C is 0.5 mol/L. The equilibrium constant  $K_c$  for the reaction is:

- (1) 0.073 (2) 0.147  
 (3) 0.05 (4) 0.026

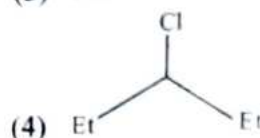
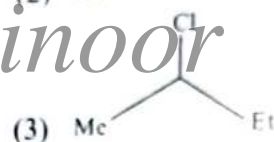
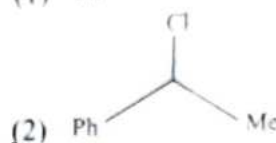
75. Given below are two statements:

**Statement I:** Boiling point of n-pentane is higher than neopentane.

**Statement II:** With branching Van der waal force decrease therefore boiling point decreases. In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.  
 (2) Statement I is incorrect but Statement II is correct.  
 (3) Both Statement I and Statement II are correct.  
 (4) Both Statement I and Statement II are incorrect.

76. Which is most reactive for  $\text{S}_\text{N}1$  reactions.



77. Calculate the radius of 3<sup>rd</sup> excited state of  $\text{Li}^{2+}$ .

- (1)  $2.82 \text{ \AA}$  (2)  $4.52 \text{ \AA}$   
 (3)  $7.23 \text{ \AA}$  (4)  $1.75 \text{ \AA}$

78. In which of the following process entropy increases?

- A. Boiling of egg.  
 B.  $\text{H}_2\text{O}_{(\text{l})} \rightarrow \text{H}_2\text{O}_{(\text{s})}$   
 C.  $\text{Cl}_{2(\text{g})} \rightarrow 2\text{Cl}_{(\text{g})}$

Choose the **correct** answer from the options given below:

- (1) A only (2) A, B and C  
 (3) A and C only (4) A and B only

79. Buckminsterfullerene contains:
- (1) 12 six-membered rings and 18 five membered rings.
  - (2) 18 six-membered rings and 12 five membered rings.
  - (3) 16 six-membered rings and 20 five membered rings.
  - (4) 20 six-membered rings and 12-five membered rings.

80. Match List-I with List-II.

| List-I<br>(Ligand Name)         | List-II<br>(Nature of ligand) |
|---------------------------------|-------------------------------|
| (A) $\text{NO}_2^-$             | (I) Hexadentate               |
| (B) $\text{OH}^-$               | (II) Didentate                |
| (C) $\text{C}_2\text{O}_4^{2-}$ | (III) Monodentate             |
| (D) $\text{EDTA}^{4-}$          | (IV) Ambidentate ligand       |

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
  - (2) A-III, B-IV, C-II, D-I
  - (3) A-III, B-IV, C-I, D-II
  - (4) A-I, B-III, C-II, D-IV
81. Given below are two statements:
- Statement I:** Aniline is less basic than ethylamine.
- Statement II:** In aniline lone pair electron of nitrogen atom participate in the resonance.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I is correct but Statement II is incorrect.
  - (2) Statement I is incorrect but Statement II is correct.
  - (3) Both Statement I and Statement II are correct.
  - (4) Both Statement I and Statement II are incorrect.

82. The electronegativity of the following elements: H, O, F, S and Cl increase in the order:
- (1)  $\text{H} < \text{O} < \text{F} < \text{S} < \text{Cl}$
  - (2)  $\text{Cl} < \text{H} < \text{O} < \text{F} < \text{S}$
  - (3)  $\text{H} < \text{S} < \text{O} < \text{Cl} < \text{F}$
  - (4)  $\text{H} < \text{S} < \text{Cl} < \text{O} < \text{F}$

83. Which of the following contains the highest number of molecules?

- (1) 2.8 g of CO
- (2) 3.2 g of  $\text{CH}_4$
- (3) 1.7 g of  $\text{NH}_3$
- (4) 3.2 g of  $\text{SO}_2$

84. On charging the lead storage battery  $\text{PbSO}_4(\text{s})$  on anode and cathode is converted into;

- (1)  $\text{PbO}_2$  and Pb respectively
- (2) Pb and  $\text{PbO}$  respectively
- (3) Pb and  $\text{PbO}_2$  respectively
- (4)  $\text{PbO}$  and Pb respectively

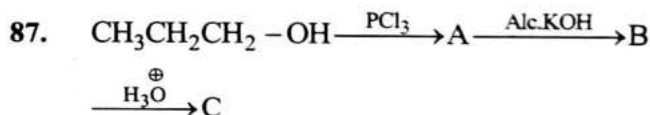
85. If 0.15 g of a solute dissolved in 15 g of solvent is boiled at a temperature higher by  $0.216^\circ\text{C}$  than that of the pure solvent. The molecular weight of the substance (molal elevation constant for the solvent is  $2.16^\circ\text{C}$ ) is:

- (1) 1.01
- (2) 10
- (3) 10.1
- (4) 100

**Chemistry : Section-B (Q. No. 86 to 100)**

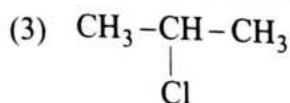
86. In a compound, A (Atomic weight = 12.5) is 25% and B (Atomic weight = 37.5) is 75% by weight. The empirical formula of the compound is:

- (1) AB
- (2)  $\text{A}_2\text{B}$
- (3)  $\text{A}_2\text{B}_2$
- (4)  $\text{A}_2\text{B}_3$



The product C is;

- (1)  $\text{CH}_3\text{CH} = \text{CH}_2$
- (2)  $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{CH}_3$



88. What would be the osmotic pressure of 0.1 M  $\text{K}_2\text{SO}_4$  solution (90% dissociated) at  $27^\circ\text{C}$ ?

(Given that value of  $R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$ )

- (1) 6.89 atm (2) 0.689 atm  
(3) 0.344 atm (4) 3.4 atm

89. Ozonolysis of 3-Methyl but-1-ene gives a mixture of:

- (1) Propanal and ethanal  
(2) Propanone and ethanal  
(3) 2-Methylpropanal and methanal  
(4) Butanone and methanal

90. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.  
**Assertion A:** Linkage isomerism arises in coordination compounds containing ambidentate Ligand.

**Reason R:** Ambidentate Ligand has two different donor atoms.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.  
(2) A is false but R is true.  
(3) Both A and R are true and R is the correct explanation of A.  
(4) Both A and R are true but R is not the correct explanation of A.

91. Compound that does not give a positive Lassaigne's test for nitrogen is:

- (1) Urea (2) Hydrazine  
(3) Aniline (4) Phenyl hydrazine

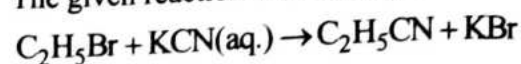
92. The dipole moment of  $\text{NH}_3$  is:

- (1) less than dipole moment of  $\text{NCl}_3$   
(2) higher than dipole moment of  $\text{NCl}_3$   
(3) equal to the dipole moment of  $\text{NCl}_3$   
(4) none of these

93. The number of moles of  $\text{KMnO}_4$  reduced by one mole of KI in alkaline medium is:

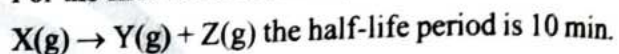
- (1) three (2) five  
(3) one (4) two

94. The given reaction is an example of:



- (1) Elimination reaction  
(2) Nucleophilic substitution  
(3) Electrophilic substitution  
(4) nucleophilic addition

95. For the first order reaction,



the half-life period is 10 min. In what period of time would the concentration of X be reduced to 10% of original concentration?

- (1) 20 min (2) 33 min  
(3) 15 min (4) 25 min

96. The pH of a 0.1 M formic acid, which is 0.1% dissociated is equal to 4. What will be the pH of another weak monobasic acid (same concentration) which is 1% dissociated?

- (1) 2 (2) 3  
(3) 1 (4) 4

97. One mole of an ideal gas at 300 K is expanded isothermally from an initial volume of 1 litre to 10 litre. The  $\Delta U$  for the process is:

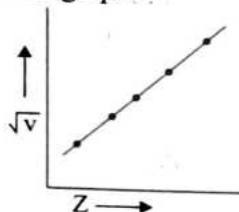
( $R = 2 \text{ Cal K}^{-1} \text{ mol}^{-1}$ )

- (1) 163.7 Cal (2) 1381.1 Cal  
(3) 9 L-atm (4) Zero

98. When 9.65 coulombs of electricity is passed through a solution of silver nitrate (atomic weight of Ag = 108) the amount of silver deposited is:

- (1) 10.8 mg (2) 5.4 mg  
(3) 16.2 mg (4) 21.2 mg

99. According to Henry Moseley, what does  $Z$  represent in the graph?



- (1) Atomic mass (2) Atomic number  
(3) Mass number (4) Number of neutrons
100. Which of the following ion is paramagnetic?  
(1)  $Ce^{+4}$  (2)  $Lu^{+3}$   
(3)  $Yb^{+2}$  (4)  $Eu^{+2}$

**Botany : Section-A (Q. No. 101 to 135)**

101. Read the following statement and select the correct option.  
(1) In prokaryotes, ribosomes are associated with the plasmid.  
(2) Ribosomes are the site of protein synthesis.  
(3) Single ribosome when gets attached to many chromosomes it is called polysome.  
(4) The ribosomes of a polysome translate the DNA into proteins.
102. Select the sequence that correctly orders the taxonomic categories of the housefly from broadest to most specific:  
(1) Musca, Muscidae, Diptera, Insecta, Arthropoda  
(2) Arthropoda, Insecta, Diptera, Muscidae, Musca  
(3) Diptera, Muscidae, Musca, Insecta, Arthropoda  
(4) Muscidae, Musca, Diptera, Insecta, Arthropoda
103. The plant body is attached to the substratum by a holdfast, and has a stalk, the stipe and leaf like photosynthetic organ – the frond in which of the following organisms?  
(1) *Gracilaria* (2) *Porphyra*  
(3) *Chara* (4) *Laminaria*

104. Read the following statements.

- A. A flower may be trimerous, tetramerous or pentamerous when the floral appendages are in multiples of 3, 4 or 5.  
B. Bracts are reduced leaves, at the axil of which a flower is borne.  
C. A flower having either only stamens or only carpels is monoecious.  
D. In cymose inflorescence, the main axis terminates in a flower, hence the growth is unlimited.  
E. Corolla may assist pollination.

How many of the above statements are correct?

- (1) Three (2) Two  
(3) One (4) Four

105. Match List-I with List-II.

| List-I                                             | List-II                    |
|----------------------------------------------------|----------------------------|
| (A) Forms nodule in the roots of leguminous plants | (I) <i>Azotobacter</i>     |
| (B) Biofertilizer in paddy field.                  | (II) <i>Glomus</i>         |
| (C) Mycorrhizra                                    | (III) <i>Cyanobacteria</i> |
| (D) Free living nitrogen fixation                  | (IV) <i>Rhizobium</i>      |

Choose the correct answer from the option given below:

- (1) A-I, B-III, C-IV, D-II  
(2) A-IV, B-III, C-II, D-I  
(3) A-II, B-IV, C-I, D-III  
(4) A-III, B-I, C-II, D-IV
106. In a DNA, nucleotide chain has AGCTTCGA sequence. The complementary nucleotide sequence of other chain would be:  
(1) TCGAAGCT (2) GCTAAGCT  
(3) TAGCATAT (4) GATCCTAG
107. What is the genetic similarity between geitonogamy and autogamy?  
(1) Functionally, they shows are self-pollination but genetically they are different  
(2) They share genes from different plant species.

- (3) Genetically, they are similar since pollen comes from the same plant.  
 (4) They have a high degree of genetic variability.

108. Match List-I with List-II.

| List-I<br>(Type of chromosome) | List-II<br>(Position of centromere) |
|--------------------------------|-------------------------------------|
| (A) Metacentric                | (I) Terminal                        |
| (B) Sub-metacentric            | (II) Middle                         |
| (C) Acrocentric                | (III) Slightly away from the middle |
| (D) Telocentric                | (IV) Close to the end               |

Choose the **correct** answer from the options given below:

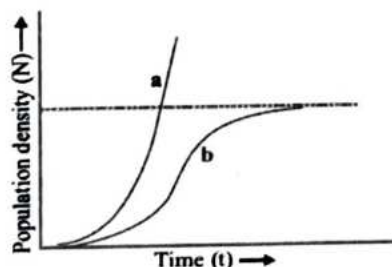
- (1) A-II, B-IV, C-I, D-III  
 (2) A-II, B-III, C-IV, D-I  
 (3) A-II, B-III, C-I, D-IV  
 (4) A-II, B-IV, C-III, D-I

109. Identify A and B in the given diagram of blue green algae. Select the **correct** option.



- (1) A – Heterocysts, B – Mucilagenous sheath  
 (2) A – Mucilagenous sheath, B – Heterocysts  
 (3) A – Heterocysts, B – Pellicle  
 (4) A – Pellicle, B – Heterocysts

110. Select the **incorrect** regarding population growth curve.



- A. Growth curve is a-when responses are not limiting the growth, plot is exponential.  
 B. Growth curve is b when responses are limiting the growth, plot is logistic.  
 C. K represents the carrying capacity, which is the horizontal line in the graph where the population density (curve a) stabilises.  
 D. r represents the intrinsic rate of increase, which is reflected in the steepness of the initial portion of the exponential growth curve (curve b)

Choose the most appropriate answer from the options given below:

- (1) A, B and C only  
 (2) C and D only  
 (3) A, B, C and D  
 (4) B and C only

111. Match List-I with List-II.

| List-I                    | List-II                                                 |
|---------------------------|---------------------------------------------------------|
| (A) Anton Von Leeuwenhoek | (I) All plants are composed of different kinds of cell  |
| (B) Robert Brown          | (II) The presence of cell wall is unique to plant cells |
| (C) Matthias Schleiden    | (III) First saw and described a living cell             |
| (D) Theodore Schwann      | (IV) Discovered the nucleus                             |

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II  
 (2) A-IV, B-III, C-I, D-II  
 (3) A-I, B-IV, C-II, D-III  
 (4) A-II, B-IV, C-I, D-III

112. The female reproductive organ in the gynoecium typically includes which parts?

- (1) Stamens and carpels  
 (2) Anther and filament  
 (3) Sepal and petals  
 (4) Stigma, style and ovary

113. The placenta forms a ridge along the ventral suture of the ovary and the ovules are borne on this ridge forming two rows. The above statement refers to:

- (1) basal placentation.
- (2) marginal placentation.
- (3) parietal placentation.
- (4) axile placentation.

114. Identify the statement that correctly matches the labels with their functional or structural attributes as given below:



- (a) Thin structures that aid in water and nutrient absorption from the soil.
- (b) The protective outer layer of cells.
- (c) Consists mainly of parenchyma cells and provides storage.
- (d) Has a casparian strip, which acts as a barrier to regulate the transfer of substances.
- (e) Lies just inside the endodermis and gives rise to lateral roots.

- (1) (a)-5, (b)-1, (c)-4, (d)-3, (e)-2
- (2) (a)-1, (b)-2, (c)-3, (d)-4, (e)-5
- (3) (a)-4, (b)-5, (c)-1, (d)-6, (e)-7
- (4) (a)-9, (b)-4, (c)-2, (d)-1, (e)-8

115. Select the correct statement about mendelian characters?

- (1) Constricted pods are dominant over inflated.
- (2) Green pods are recessive over yellow pods.
- (3) Tall plants are dominant over dwarf plants.
- (4) White flowers are dominant over violet flowers.

116. In Mendel's experiments on pea plants, the trait that disappeared in the  $F_1$  generation but reappeared in the  $F_2$  generation is called:

- (1) dominant trait.
- (2) recessive trait.
- (3) co-dominant trait.
- (4) polygenic trait.

117. Given below are two statements:

**Statement I:** Microbodies are membrane bound minute vesicles that contain various enzymes.

**Statement II:** Microbodies are absent in animal cells.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

118. Select the **incorrect** statement.

- (1) Citrus canker is caused by virus.
- (2) Dinoflagellates, euglenoids and slime mould are placed under kingdom protista.
- (3) Members of kingdom protista are primarily aquatic.
- (4) In slime moulds, the spores are dispersed by air currents.

119. Sexual spores are produced exogenously in:

- (1) *Alternaria* and *Trichoderma*.
- (2) *Puccinia* and *Ustilago*.
- (3) *Penicillium* and *Aspergillus*.
- (4) *Neurospora* and *Colletotrichum*.

120. Match List-I and List-II.

| List-I<br>(Protozoans) | List-II<br>(Example)     |
|------------------------|--------------------------|
| (A) Amoeboid           | (I) <i>Entamoeba</i>     |
| (B) Flagellated        | (II) <i>Plasmodium</i>   |
| (C) Ciliated           | (III) <i>Trypanosoma</i> |
| (D) Sporozoans         | (IV) <i>Paramoecium</i>  |

Choose the **correct** answer from the option given below:

- (1) A-I, B-II, C-III, D-IV
- (2) A-II, B-I, C-III, D-IV
- (3) A-I, B-III, C-IV, D-II
- (4) A-IV, B-III, C-II, D-I

121. Ovary is one chambered but it becomes two-chambered due to the formation of false septum in:

- (1) tomato and lemon.
- (2) mustard and *Argemone*.
- (3) *Dianthus* and *Primrose*.
- (4) sunflower and marigold.

122. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** Plants capture only 2-10 per cent of the PAR and this small amount of energy sustains the entire living world.

**Reason R:** Only 10 per cent of the energy is transferred to each trophic level from the lower trophic level.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

123. The discontinuously synthesised fragment in DNA replication are later joined by the enzyme:

- (1) DNA ligase.
- (2) DNA dependent DNA polymerase
- (3) permease.
- (4) DNA dependent RNA polymerase.

124. Identify the **correct** statements from the following.

- A. Cell-plate formed during cytokinesis represents the middle lamella.
- B. The growth of multicellular organisms is due to mitosis.
- C. Mitosis usually results in the production of diploid daughter cells with different genetic complements.
- D. A very significant contribution of mitosis is cell repair.

Choose the **most appropriate** answer from the options given below:

- (1) B and D only
- (2) A, B and D
- (3) A and C only
- (4) A, B, C and D

125. Identify the given image A and B structure of a leaf which represents venation. Select the correct option.





- (1) A - Reticulate - monocotyledons plants,  
B - Parallel-dicotyledonous plants.
- (2) A - Reticulate - dicotyledonous plants,  
B - Parallel-monocotyledons plants.
- (3) A - Parallel - dicotyledonous plants,  
B - Reticulate monocotyledons plants.
- (4) A - Reticulate-monocotyledons plants.  
B - Parallel-dicotyledonous plants.

126. Match List-I and List-II.

| List-I                             | List-II                 |
|------------------------------------|-------------------------|
| (A) Sacred groves                  | (I) Western Ghat region |
| (B) Botanical gardens              | (II) <i>Ex-situ</i>     |
| (C) Narrowly utilitarian arguments | (III) <i>In-situ</i>    |
| (D) Hotspot                        | (IV) Economic benefit   |

Choose the **correct** answer from the option given below:

- (1) A-III, B-II, C-IV, D-I  
(2) A-I, B-II, C-III, D-IV  
(3) A-I, B-III, C-II, D-IV  
(4) A-IV, B-I, C-II, D-III

127. In relation to gross primary productivity and net primary productivity of an ecosystem, which one of the following statements is **correct**?

- (1) Gross primary productivity is always more than net primary productivity.
- (2) Gross primary productivity and net primary productivity are one and same.
- (3) There is no relationship between Gross primary productivity and net primary productivity.
- (4) Gross primary productivity is always less than net primary productivity.

128. Read the given statements.

**Statement I:** In an aquatic ecosystem, grazing food chain (GFC) is the major conduit for energy flow.

**Statement II:** Pyramid of energy is always upright, can never be inverted.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

129. Match List-I and List-II.

| List-I<br>(No. of Organisms)     | List-II<br>(No. of Base pairs) |
|----------------------------------|--------------------------------|
| (A) Bacteriophage lamda          | (I) $3.3 \times 10^9$ bp       |
| (B) <i>E.coli</i>                | (II) $4.6 \times 10^6$ bp      |
| (C) Haploid content of human DNA | (III) 48502 bp                 |
| (D) Bacteriophage                | (IV) 5386 nucleotides          |

Choose the **correct** answer from the option given below:

- (1) A-I, B-II, C-III, D-IV  
(2) A-III, B-II, C-I, D-IV  
(3) A-I, B-III, C-II, D-IV  
(4) A-III, B-I, C-II, D-IV

130. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** Mitosis accomplishes only the segregation of duplicated chromosomes into daughter nuclei.

**Reason R:** During mitosis, the cell itself is divided into two daughter cells by the separation of cytoplasm at the end of which cell division gets completed.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is correct but R is not correct.
- (2) A is not correct but R is correct.
- (3) Both A and R are correct and R is the correct explanation of A.
- (4) Both A and R are correct but R is NOT the correct explanation of A.

131. Read the following statements.

- A. The cells of the human cheek have an outer membrane as the delimiting structure of the cell.
- B. Cells that have membrane bound nuclei are called eukaryotic whereas cells that lack a membrane bound nucleus are prokaryotic.
- C. In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the nucleus.
- D. The nucleus is the main arena of cellular activities in both the plant and animal cells.

Choose the **incorrect** option.

- (1) C and D only
- (2) A, C and B only
- (3) B, D, and A only
- (4) A and C only

132. Which of the following is **NOT** related to the light reaction?

- (1) Also called photochemical phase
- (2) Light absorption
- (3) Electron transport
- (4) Water splitting

133. Given below are two statements:

**Statement I:** Genetics is a branch of biology which deals with principles of inheritance and its practices.

**Statement II:** Variation is the degree by which progeny differ from their parents.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

134. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R: **Assertion A:** The presence of introns is reminiscent of antiquity.

**Reason R:** The process of splicing represents the dominance of DNA-world.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

135. Which of the following statements is **incorrect**?

- (1) Stroma lamellae have PS I only and lack NADP reductase.
- (2) Grana lamellae have both PS I and PS II.
- (3) Cyclic photophosphorylation involves both PS I and PS II.
- (4) Both ATP and NADPH +  $H^+$  are synthesized during non-cyclic photophosphorylation.

**Botany : Section-B (Q. No. 136 to 150)**

136. Given below are two statements.

**Statement I:** Interphase is the time during which the cell is preparing for division by undergoing both cell growth and DNA replication in an orderly manner.

**Statement II:** During  $G_1$  phase, the cell is metabolically active and continuously grows but does not replicate its DNA.



In the light of above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

137. Select the **correct** statements regarding Thomas Hunt Morgan.

- A. He explained genetic maps.
- B. He coined the term linkage.
- C. He gave experimental verification of the chromosomal theory of inheritance.
- D. He studied genes that were autosome linked.

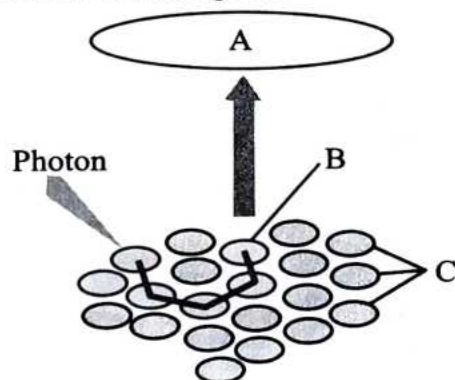
Choose the most appropriate answer from the options given below:

- (1) A and C only
- (2) B and C only
- (3) C and D only
- (4) A and B only

138. Starch sheath is present in:

- (1) monocotyledonous root.
- (2) dicotyledonous root.
- (3) monocotyledonous stem.
- (4) dicotyledonous stem.

139. Given figure depicts the Light Harvesting Complex (LHC) of Photosystem-I (PS-I) and select the **correct** option.



|     | A                 | B                 | C                 |
|-----|-------------------|-------------------|-------------------|
| (1) | Core molecules    | Plastocyanin      | Primary acceptor  |
| (2) | Primary acceptor  | Reaction centre   | Pigment molecules |
| (3) | Reaction centre   | Pigment molecules | Primary Acceptor  |
| (4) | Pigment molecules | Primary acceptor  | Reaction Centre   |

140. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** In RNA the uracil is found at the place of thymine.

**Reason R:** In RNA, nucleotide residue has no additional  $-OH$  group present at 2'-position in the Ribose Sugar.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is correct but R is not correct.
- (2) A is not correct but R is correct.
- (3) Both A and R are correct and R is the correct explanation of A.
- (4) Both A and R are correct but R is NOT the correct explanation of A.

141. Choose the **correct** option regarding fermentation.

- (1) Found in all bacteria and yeast.
- (2) Lactic acid is the end product in some organisms.
- (3) Production of methyl alcohol.
- (4)  $NADH_2$  is not produced.

142. Given below are two statements:

**Statement I:** Inheritance is the process by which characters are passed on from parent to progeny.

**Statement II:** Variation is the degree by which progeny differ from their parents.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

143. Which type of aestivation is characteristic of family Malvaceae?

- |             |               |
|-------------|---------------|
| (1) Valvate | (2) Imbricate |
| (3) Twisted | (4) Vexillary |

144. Match List-I with List-II.

| List-I                                                   | List-II           |
|----------------------------------------------------------|-------------------|
| (A) Test cross<br>genotypic ratio in<br>monohybrid cross | (I) 1 : 1 : 1 : 1 |
| (B) Phenotypic ratio<br>of monohybrid<br>cross           | (II) 1 : 1        |
| (C) Incomplete<br>dominance<br>genotypic ratio           | (III) 1 : 2 : 1   |
| (D) Test cross<br>phenotypic ratio<br>in dihybrid cross  | (IV) 3 : 1        |

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-III, D-I
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-II, C-I, D-IV
- (4) A-II, B-I, C-IV, D-III

145. Select the **correct** statement.

- (1) von Tschermak coined the term linkage.
- (2) Punnett square was developed by a British scientist.
- (3) Gregor Mendel noted that the behaviour of chromosomes was parallel to the behaviour of genes.
- (4) Law of segregation was given by Thomas Hunt Morgan.

146. Select the **mismatch** from the following.

- |                         |                                 |
|-------------------------|---------------------------------|
| (1) Cytoskeleton        | Maintain the shape of cell.     |
| (2) SER                 | Steroid synthesis.              |
| (3) 70S ribosome        | Found in prokaryotic cell only. |
| (4) Contractile vacuole | Helps in excretion.             |

147. Which of the following tissues are not the product of dedifferentiation?

- (a) Interfascicular cambium
  - (b) Tracheary elements
  - (c) Cork cambium
  - (d) Cork cells
  - (e) Secondary xylem and phloem elements
- Options:

- |                   |                   |
|-------------------|-------------------|
| (1) (a), (b), (c) | (2) (c), (d), (e) |
| (3) (b), (d), (e) | (4) (b), (c), (d) |

148. The highest number of species of major taxa of plants in the world is represented by:

- |           |             |
|-----------|-------------|
| (1) algae | (2) lichens |
| (3) fungi | (4) mosses  |

149. Match List-I with List-II.

| List-I                           | List-II             |
|----------------------------------|---------------------|
| (A) Cristae                      | (I) Golgi apparatus |
| (B) Cisternae                    | (II) Cilia          |
| (C) Flattened<br>membranous sacs | (III) Mitochondria  |
| (D) Axoneme                      | (IV) Thylakoids     |

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-I, C-IV, D-II
- (4) A-IV, B-I, C-III, D-II

150. Read the given statements.

**Statement I:** Competition occurs when closely related species compete for the same resources that are limiting.

**Statement II:** Predators help in maintaining species diversity in a community.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement-I is correct but Statement-II is incorrect.
- (2) Statement-I is incorrect but Statement-II is correct.
- (3) Both Statement-I and Statement-II are correct.
- (4) Both Statement-I and Statement-II are incorrect.

**Zoology : Section-A (Q. No. 151 to 185)**

151. Which part of the tobacco plant is infected by *Meloidogyne incognita*?

- (1) Stem
- (2) Root
- (3) Flower
- (4) Leaf

152. Match List-I with List-II.

| List-I                           | List-II          |
|----------------------------------|------------------|
| (A) Inspiratory capacity         | (I) TV+ERV       |
| (B) Expiratory capacity          | (II) ERV+RV      |
| (C) Functional residual capacity | (III) ERV+TV+IRV |
| (D) Vital capacity               | (IV) TV+IRV      |

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-II, D-III
- (2) A-I, B-II, C-IV, D-III
- (3) A-III, B-II, C-IV, D-I
- (4) A-II, B-IV, C-I, D-III

153. Neutrophils and macrophages cells in the human body exhibit which type of movement?

- (1) Amoeboid movement
- (2) Ciliary movement
- (3) Flagellar movement
- (4) Muscular movement

154. Who proposed the theory of use and disuse of organs?

- (1) Lamarck
- (2) Hugo de Vries
- (3) Charles Darwin
- (4) Alfred Wallace

155. Match List-I with List-II.

| List-I<br>(Elements) | List-II<br>(% Weight of Earth's crust) |
|----------------------|----------------------------------------|
| (A) Oxygen           | (I) 27.7                               |
| (B) Silicon          | (II) 0.03                              |
| (C) Sulphur          | (III) 3.6                              |
| (D) Calcium          | (IV) 46.6                              |

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-IV, D-II
- (2) A-IV, B-I, C-II, D-III
- (3) A-III, B-II, C-I, D-IV
- (4) A-II, B-IV, C-I, D-III

156. The circulatory system of *Rana tigrina* consists of:

- A. a closed type.
  - B. heart, blood vessels, blood, lymph, lymph channels and lymph nodes.
  - C. a triangular sinus venosus.
  - D. double membranous pleura.
- (1) A, B and C only
  - (2) B, C and D only
  - (3) C and D only
  - (4) A, B and D only

157. The process of introducing recombinant DNA into bacterial cells is called:
- microinjection.
  - transformation.
  - spooling.
  - elution.

158. Consider the following features:

- organ system level of organisation.
- triploblastic animals.
- true coelomates with segmentation of the body.

Select the **correct** option for phyla which possess all the above characteristics.

- Annelida, Cnidaria and Chordata
- Annelida, Arthropoda and Chordata
- Annelida, Arthropoda and Cnidaria
- Arthropoda, Cnidaria and Chordata

159. Given below are two statements:

**Statement I:** The Big Bang theory attempts to explain to us the origin of the universe.

**Statement II:** Formation of life was preceded by chemical evolution.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

160. Match List-I with List-II.

| List-I                      | List-II                   |
|-----------------------------|---------------------------|
| (A) Male accessory gland    | (I) Testis                |
| (B) Covering of glans penis | (II) Penis                |
| (C) Male external genitalia | (III) Bulbourethral gland |
| (D) Primary sex organ       | (IV) Foreskin             |

Choose the **correct** answer from the options given below:

- A-I, B-III, C-IV, D-II
- A-I, B-II, C-IV, D-III
- A-III, B-IV, C-II, D-I
- A-II, B-IV, C-I, D-III

161. Insulin is:

- peptide hormone.
- steroid hormone.
- thyroid hormone.
- Catecholamine.

162. Given below are two statements:

**Statement I:** *cryIIAb* controls corn borer.

**Statement II:** Toxicity testing in transgenic animals allow us to obtain results in less time.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

163. Match List-I with List-II.

| List-I                 | List-II                |
|------------------------|------------------------|
| (A) Bacterial disease  | (I) Ringworm infection |
| (B) Fungal disease     | (II) Filariasis        |
| (C) Viral disease      | (III) Typhoid          |
| (D) Helminthic disease | (IV) Common cold       |

Choose the **correct** answer from the options given below:

- A-III, B-I, C-IV, D-II
- A-II, B-I, C-IV, D-III
- A-I, B-IV, C-II, D-III
- A-III, B-IV, C-II, D-I

164. Which of the following statements are correct?
- Paper is cellulosic in nature.
  - A nucleotide has a heterocyclic compound, a monosaccharide and a phosphoric acid.
  - Inulin is a heteropolymer of chitin.
  - Essential amino acids are not needed to be supplied through our diet.
- A and B
  - B and C
  - C and D
  - A and C

165. In the context of amniocentesis, which of the following statements is incorrect?
- It can be used for the detection of Down syndrome.
  - It can be used for the detection of sickle cell anemia.
  - Some of the amniotic fluid of the developing foetus is taken to analyse the fetal cells and dissolved substances.
  - It is legal for prenatal sex determination.

166. Match List-I with List-II.

| List-I             | List-II                           |
|--------------------|-----------------------------------|
| (A) <i>Hind-II</i> | (I) Recognition sequence          |
| (B) pBR322         | (II) Restriction enzyme           |
| (C) GAATTC         | (III) Antibiotic resistance genes |
| (D) $\text{amp}^R$ | (IV) Cloning vector               |

Choose the correct answer from the options given below:

- A-III, B-I, C-IV, D-II
- A-II, B-IV, C-I, D-III
- A-I, B-III, C-II, D-IV
- A-III, B-IV, C-II, D-I

167. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** Tissue matching is essential before undertaking any graft/transplant.

**Reason R:** The humoral-mediated immune response is responsible for the graft rejection. In the light of the above statements, choose the correct answer from the options given below:

- A is true but R is false.
- A is false but R is true.
- Both A and R are true and R is the correct explanation of A.
- Both A and R are true but R is NOT the correct explanation of A.

168. All of the given pairs of adult animals have bilateral symmetry, **except**:

- Fasciola* and *Wuchereria*.
- Hirudinaria* and *Locusta*.
- Loligo* and *Saccoglossus*.
- Antedon* and *Aurelia*.

169. Opening of the "X" is often covered partially by a membrane called "Y". The "Y" is often torn during the first coitus (intercourse). Identify "X" and "Y".

| X          | Y           |
|------------|-------------|
| (1) Uterus | Hymen       |
| (2) Vagina | Birth canal |
| (3) Vagina | Hymen       |
| (4) Cervix | Hymen       |

170. Match List-I with List-II.

| List-I              | List-II                  |
|---------------------|--------------------------|
| (A) Pituitary gland | (I) Grave's disease      |
| (B) Thyroid gland   | (II) Diabetes mellitus   |
| (C) Adrenal gland   | (III) Diabetes insipidus |
| (D) Pancreas        | (IV) Addison's disease   |

Choose the correct answer from the options given below:

- A-I, B-III, C-II, D-IV
- A-III, B-I, C-IV, D-II
- A-IV, B-II, C-I, D-III
- A-I, B-III, C-IV, D-II

171. The lymphoid tissue located within the lining of major tracts, constitutes about \_\_\_\_\_ of the lymphoid tissue in the body.

Choose the option that fills the blank correctly.

- (1) 50% (2) 70%  
(3) 90% (4) 20%

172. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

**Assertion A:** *Taq* polymerase is used in polymerase chain reaction.

**Reason R:** It is a heat-stable enzyme.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.  
(2) A is false but R is true.  
(3) Both A and R are true and R is the correct explanation of A.  
(4) Both A and R are true but R is NOT the correct explanation of A.

173. How many of the following amino acids are aromatic?

Tryptophan, Glutamic acid, Tyrosine, Lysine, Valine

- (1) Two (2) One  
(3) Four (4) Three

174. Match List-I with List-II.

| List-I                       | List-II                                           |
|------------------------------|---------------------------------------------------|
| (A) Inter-ventricular septum | (I) Lower left corner of the right atrium         |
| (B) Inter-atrial septum      | (II) Right upper corner of the right atrium       |
| (C) Sino-atrial node         | (III) Separates the left and the right ventricles |
| (D) Atrio-ventricular node   | (IV) Separates the right and the left atria       |

Choose the correct answer from the options given below:

- (1) A-IV, B-III C-I, D-II  
(2) A-III, B-IV, C-I, D-II  
(3) A-III, B-IV, C-II, D-I  
(4) A-IV, B-III, C-II, D-I

175. In humans, the total thickness of the diffusion membrane of alveolus capillary is:

- (1) much more than a centimetre.  
(2) much less than a millimetre.  
(3) equal to a millimetre.  
(4) equal to a centimetre.

176. Neurons can:

- (1) detect different kinds of stimuli only.  
(2) transmit different kinds of stimuli only.  
(3) receive different kinds of stimuli only.  
(4) All of these

177. Match List-I with List-II.

| List-I           | List-II                                 |
|------------------|-----------------------------------------|
| (A) Mendel       | (I) Embryological support for evolution |
| (B) Hugo deVries | (II) Chemical evolution                 |
| (C) Ernst Heckel | (III) Inheritable factors               |
| (D) Oparin       | (IV) Mutation                           |

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II  
(2) A-II, B-III, C-IV, D-I  
(3) A-IV, B-II, C-I, D-III  
(4) A-II, B-IV, C-I, D-III

178. Given below are two statements:

**Statement I:** Leydig cells are located in interstitial spaces.

**Statement II:** The seminal plasma lacks calcium. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.  
 (2) Statement I is incorrect but Statement II is correct.  
 (3) Both Statement I and Statement II are correct.  
 (4) Both Statement I and Statement II are incorrect.

179. The plasmid:

- (1) is an autonomously replicating circular extra-chromosomal DNA present in bacterial cells.  
 (2) is a component of the cell wall of bacteria.  
 (3) is found inside the nucleus.  
 (4) helps in respiration.

180. The compound that is **not** found in the acid-insoluble pool is:

- (1) protein. (2) nucleic acid.  
 (3) glycine. (4) polysaccharide.

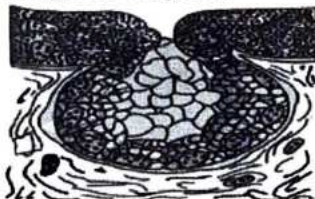
181. Match List-I with List-II.

| List-I    | List-II              |
|-----------|----------------------|
| (A) Lungs | (I) Biliverdin       |
| (B) Liver | (II) Waxes           |
| (C) Sweat | (III) Lactic acid    |
| (D) Sebum | (IV) CO <sub>2</sub> |

Choose the **correct** answer from the options given below:

- (1) A-IV, B-I, C-III, D-II  
 (2) A-II, B-I, C-IV, D-III  
 (3) A-II, B-III, C-IV, D-I  
 (4) A-IV, B-II, C-III, D-I

182. Identify the given diagram.



- (1) Multicellular glandular epithelium  
 (2) Unicellular gland  
 (3) Goblet cells  
 (4) Squamous epithelium

183. In the nephron, electrolytes and water absorption from the glomerular filtrate are at their maximum in:

- (1) proximal convoluted tubule.  
 (2) loop of Henle.  
 (3) glomerulus.  
 (4) distal convoluted tubule.

184. Match List-I with List-II.

| List-I                             | List-II                          |
|------------------------------------|----------------------------------|
| (A) First movements of the foetus  | (I) By the end of third month    |
| (B) Eye-lids separate              | (II) By the end of fifth month   |
| (C) Limbs and digits formed        | (III) By the end of second month |
| (D) External genital organs formed | (IV) By the end of sixth month   |

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV  
 (2) A-II, B-III, C-I, D-IV  
 (3) A-II, B-IV, C-III, D-I  
 (4) A-IV, B-II, C-III, D-I

185. Select the option with the **correct** features for red muscle fibres.

- A. Few mitochondria  
 B. High myoglobin content  
 C. Aerobic muscles  
 D. Plenty of sarcoplasmic reticulum  
 (1) A and B only (2) B and C only  
 (3) A, B and C (4) A, C and D

**Zoology : Section-B (Q. No. 186 to 200)**

186. Which of the following statements is **correct** with regard to the principle of safe blood transfusion?

- (1) The donor's red blood corpuscles should not contain antibodies against the recipient's blood.  
 (2) The recipient's blood should not contain antigens against the donor's antibodies.

- (3) The recipient's red blood corpuscles should not contain antibodies against the donor's antigen.
- (4) The recipient's blood should not contain antibodies against the red blood corpuscles of the donor.

187. Match List-I with List-II.

| List-I          | List-II             |
|-----------------|---------------------|
| (A) Golden rice | (I) Tissue culture  |
| (B) Explant     | (II) Vitamin 'A'    |
| (C) Eli Lilly   | (III) Free of virus |
| (D) Meristem    | (IV) Insulin        |

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-IV, D-III
- (2) A-IV, B-III, C-II, D-I
- (3) A-I, B-II, C-III, D-IV
- (4) A-II, B-I, C-IV, D-III

188. Given below are some statements:

- A. Cancer causing viruses have genes called viral oncogenes.
- B. Malignant tumors remain confined to their original location.
- C. Cancer cells do not exhibit contact inhibition.
- D. X-rays and UV rays are not potent carcinogens.

Which of the above statements are **incorrect** regarding cancer?

- (1) A and B                      (2) B and D
- (3) A and C                      (4) C and D

189. Match List-I with List-II.

| List-I                  | List-II                  |
|-------------------------|--------------------------|
| (A) Periodic abstinence | (I) Intrauterine Devices |
| (B) Cervical caps       | (II) Traditional method  |
| (C) Multiload 375       | (III) Barrier method     |
| (D) Tubectomy           | (IV) Surgical method     |

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-I, B-IV, C-II, D-III
- (4) A-II, B-III, C-I, D-IV

190. Given below are two statements:

**Statement I:** Adult human haemoglobin, consists of two subunits of  $\alpha$  type and two subunits of  $\beta$  type.

**Statement II:** C - terminal amino acid is present at the left end of the protein chain.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

191. Match List-I with List-II.

| List-I                    | List-II                                    |
|---------------------------|--------------------------------------------|
| (A) Ball and socket joint | (I) Between carpal and metacarpal of thumb |
| (B) Hinge joint           | (II) Between humerus and pectoral girdle   |
| (C) Pivot joint           | (III) Knee joint                           |
| (D) Saddle joint          | (IV) Between atlas and axis                |

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-II, C-IV, D-I
- (3) A-IV, B-II, C-I, D-III
- (4) A-II, B-IV, C-I, D-III

102. Read the following statements:

- The 'new' variety of Basmati had actually been derived by crossing Indian basmati with semi-dwarf varieties.
- The natural cow milk is nutritionally a more balanced product for human babies than milk produced by Rosie.
- Insulin used for diabetes was earlier extracted from pancreas of slaughtered cattle and pigs.
- Enzyme replacement therapy is a permanent cure for adenosine deaminase-deficient patients.

Which of the above statements are correct?

- A and D
- B and D
- A and C
- B and C

103. Match List-I with List-II.

| List-I                | List-II                                               |
|-----------------------|-------------------------------------------------------|
| (A) Corpus callosum   | (I) Controls body temperature.                        |
| (B) Medulla oblongata | (II) Cerebral aqueduct                                |
| (C) Hypothalamus      | (III) Control cardiovascular reflexes.                |
| (D) Midbrain          | (IV) Connect the left and right cerebral hemispheres. |

Choose the correct answer from the options given below:

- A-III, B-I, C-IV, D-II
- A-II, B-III, C-I, D-IV
- A-I, B-IV, C-III, D-II
- A-IV, B-III, C-I D-II

194. Given below are two statements:

**Statement I:** Ctenophores are exclusively marine.

**Statement II:** Ctenophores do not exhibit bioluminescence at all.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Statement I is correct but Statement II is incorrect.
- Statement I is incorrect but Statement II is correct.
- Both Statement I and Statement II are correct.
- Both Statement I and Statement II are incorrect.

195. Match List-I with List-II.

| List-I                    | List-II                                                                |
|---------------------------|------------------------------------------------------------------------|
| (A) Origin of replication | (I) Gel electrophoresis                                                |
| (B) Selectable marker     | (II) Remove nucleotides from the ends of the DNA.                      |
| (C) Exonucleases          | (III) Control the copy number.                                         |
| (D) Ethidium bromide      | (IV) Identifying and eliminating non-transformants from transformants. |

Choose the correct answer from the options given below:

- A-IV, B-III, C-I, D-II
- A-I, B-II, C-IV, D-III
- A-II, B-III, C-IV, D-I
- A-III, B-IV, C-II, D-I

196. Select the option representing correct examples of analogous organs.

- Sweet potato and potato.
  - Vertebrate hearts and brains.
  - Thorn of *Bougainvillea* and tendril of *Cucurbita*.
  - Eyes of the *Octopus* and mammals.
- A and D only
  - A, B and C
  - B, C and D
  - C and D only

197. Match List-I with List-II.

| List-I                                  |       | List-II          |
|-----------------------------------------|-------|------------------|
| (A) Boat shaped 7 <sup>th</sup> sternum | (I)   | 10 segments      |
| (B) Abdomen                             | (II)  | Female cockroach |
| (C) Anal styles                         | (III) | 3 segments       |
| (D) Thorax                              | (IV)  | Male cockroach   |

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-III, D-IV
- (2) A-I, B-II, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-I, C-IV, D-III

198. Match List-I with List-II.

| List-I        |       | List-II        |
|---------------|-------|----------------|
| (A) Pigments  | (I)   | Abrin          |
| (B) Toxins    | (II)  | Concanavalin A |
| (C) Alkaloids | (III) | Carotenoids    |
| (D) Lectins   | (IV)  | Morphine       |

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-I, B-II, C-IV, D-III
- (3) A-II, B-III, C-IV, D-I
- (4) A-IV, B-III, C-II, D-I

199. Given below are some statements:

- A. The acrosome of the sperm produces energy for the movement of the tail.
- B. The sperm head contains an elongated diploid nucleus.
- C. The zona pellucida layer blocks the entry of additional sperms.
- D. The myometrium undergoes cyclical changes during menstrual cycle.

Choose the **incorrect** statements from the options given below:

- (1) A and D only
- (2) B and C only
- (3) A, B and D
- (4) B, C and D

200. Match List-I with List-II.

| List-I              |       | List-II  |
|---------------------|-------|----------|
| (A) <i>Macropus</i> | (I)   | Amphibia |
| (B) <i>Columba</i>  | (II)  | Reptilia |
| (C) <i>Galotes</i>  | (III) | Aves     |
| (D) <i>Hyla</i>     | (IV)  | Mammalia |

Choose the **correct** answer from the options given below:

- (1) A-I, B-III, C-II, D-IV
- (2) A-II, B-III, C-IV, D-I
- (3) A-IV, B-II, C-I, D-III
- (4) A-IV, B-III, C-II, D-I

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